

STAT 314500 - Applied Partial Differential Equations
Assignment 4: Laplace's Equation and Wave equation

This is an individual assignment. You may collaborate with others but are expected to submit your own work. Show all relevant work. Incomplete work will be docked points.

Note: if a question is marked with an asterisk, *, and appears vague - that's on purpose. Questions marked * are conceptual short-answer questions. Do your best to reason these through, and describe that reasoning concisely. We are looking more for reasoning than precision.

Reading: Chapters 2, 3 and 4 in Applied Partial Differential Equations with Fourier Series and Boundary Value Problems (Richard Haberman).

Question 1 Consider the Laplace's equation inside a rectangle $\Delta u = 0, (x, y) \in (0, L_1) \times (0, L_2)$ with boundary conditions

$$\partial_x u(0, y) = \partial_y(x, 0) = \partial_x u(L_1, y) = 0, \quad \partial_y u(x, L_2) = f(x).$$

1. Briefly explain when there is a solution to this problem from a physical point of view. (2 points)
2. Solve this problem using separation of variables. Verify that the condition under which this works is only when part (i) is satisfied. (5 points)
3. The solution in part (ii) has an arbitrary constant. Determine this constant if u was the steady state solution to a heat equation $\partial_t v = k\Delta v$ with initial condition $g(x, y)$. (3 points)

Question 2 Solve the Laplace's equation outside a circular disk centred at the origin of radius R subject to the boundary condition

$$u(R, \theta) = \log 2 + 4 \cos(3\theta),$$

where (r, θ) denotes the polar coordinate representation. (7 points)

Question 3 Solve the Laplace's equation in a circular annulus $a < r < b$ subject to the boundary condition (7 points)

$$\partial_r u(a, \theta) = 0, \quad u(b, \theta) = g(\theta).$$

Question 4 Consider the one dimensional wave equation

$$\begin{aligned} \frac{\partial^2 u}{\partial t^2} &= c^2 \frac{\partial^2 u}{\partial x^2} \\ u(t, 0) &= 0 \\ u(t, L) &= 0 \\ u(0, x) &= f(x) \\ \frac{\partial u}{\partial t}(0, x) &= g(x) \end{aligned} \tag{1}$$

1. Separation of variables yields a solution of the form

$$u(t, x) = \sum_{n=1}^{\infty} \left(A_n \cos \frac{n\pi ct}{L} + B_n \sin \frac{n\pi ct}{L} \right) \sin \frac{n\pi x}{L}.$$

Show that this is equivalent to $u(x, t) = \phi(x + ct) + \psi(x - ct)$ for some functions ϕ and ψ . (5 points)

2. Define the energy

$$E(t) = \frac{1}{2} \int_0^L \left| \frac{\partial u}{\partial t} \right|^2 dx + \frac{c^2}{2} \int_0^L \left| \frac{\partial u}{\partial x} \right|^2 dx.$$

This can be thought of as the sum of a kinetic energy and a potential energy. Show that (7 points)

$$\frac{dE}{dt} = c^2 \left[(\partial_x u \partial_t u)(x = L) - (\partial_x u \partial_t u)(x = 0) \right].$$

3. Use part (ii) to prove uniqueness of solutions to the wave equation (1). (7 points)

Question 5

Show that the following backward heat equation is not well posed: (7 points)

$$\begin{aligned} \frac{\partial u}{\partial t} &= -k \frac{\partial^2 u}{\partial x^2}, \quad k > 0 \\ u(t, 0) &= 0 \\ u(t, L) &= 0 \\ u(0, x) &= f(x) \end{aligned}$$

Hint: You can prove this by showing that if you perturb the initial condition by a small amplitude, the solution changes considerably. For example, you can see what happens to the solution if you replace $f(x)$ by $f(x) + \frac{1}{n} \sin \frac{n\pi x}{L}$. This exercise shows that the backward heat equation is unstable, thereby not well posed.